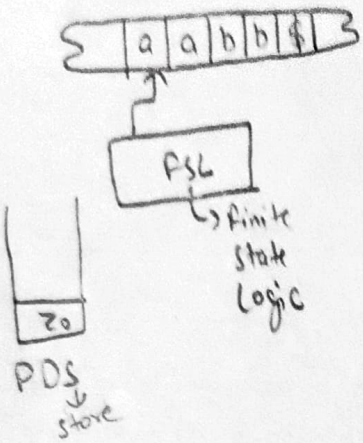
Push- Down Automata



$\delta(q_0, a, z_0) = (q_0, aa)$ inp++
State input stack — State stack

$\Rightarrow \{a^n b^n \mid n \geq 0\}$

R0 $\delta(q_0, \$, z_0) = (q_f, z_0)$

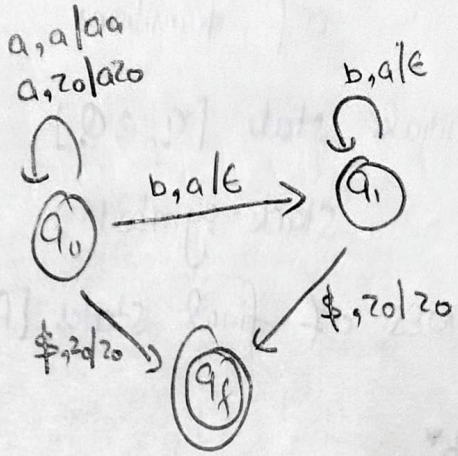
R1 $\delta(q_0, a, z_0) = (q_0, a z_0)$ (POP / PUSH)

R2 $\delta(q_0, a, a) = (q_0, aa)$

R3 $\delta(q_0, b, a) = (q_1, \epsilon)$

R4 $\delta(q_1, b, a) = (q_1, \epsilon)$

R5 $\delta(q_1, \$, z_0) = (q_f, z_0)$



$\Rightarrow \{a^n b^{n+2} \mid n \geq 0\}$

aabbbb

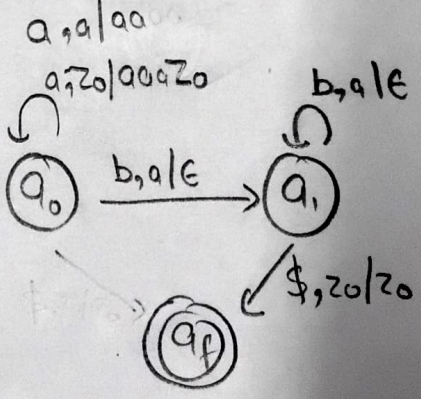
$\delta(q_0, a, z_0) = (q_0, aaaz_0)$

$\delta(q_0, a, a) = (q_0, aa)$

$\delta(q_0, b, a) = (q_1, \epsilon)$

$\delta(q_1, b, a) = (q_1, \epsilon)$

$\delta(q_1, \$, z_0) = (q_f, z_0)$



Wednesday
14 Oct, 2026

$\{a^n b^m c^n \mid n,m \geq 0\}$

PDA = $\langle Q, \Sigma, \tau, \delta, q_0, Z_0, F \rangle$

Q = FNES of all states

Σ = " " " i/p symbols

τ = " of stack symbols $[\tau \subseteq \Sigma]$

δ = " of transitions

q_0 = initial state $[q_0 \in Q]$

Z_0 = " stack symbol

F = FNES of final states $[F \subseteq Q]$

$\Rightarrow a^* b^*$

